

MITRE ATT&CK Framework for SOC Analysts

1. Introduction

The MITRE ATT&CK framework is a globally recognized cybersecurity knowledge base created by the ^[2]entity^[2]["organization", "MITRE Corporation", "U.S. nonprofit organization developing cybersecurity frameworks"]^[2]. It describes how attackers behave in real-world cyber attacks using structured tactics, techniques, and procedures (TTPs).

For SOC analysts, it is one of the most important frameworks for threat detection, hunting, and incident response.

2. What is MITRE ATT&CK?

MITRE ATT&CK is a matrix that explains:

- WHY attackers act (Tactics)
- HOW they achieve their goals (Techniques)
- REAL-world behaviors (Procedures)

Instead of focusing only on malware or IPs, it focuses on attacker behavior patterns.

3. Structure of MITRE ATT&CK

3.1 Tactics (The “Why”)

Tactics represent the attacker’s objective. Examples include:

- Initial Access
 - Execution
 - Persistence
 - Privilege Escalation
 - Defense Evasion
 - Credential Access
 - Discovery
 - Lateral Movement
 - Exfiltration
 - Impact
-

3.2 Techniques (The “How”)

Techniques describe how attackers achieve tactics. Examples:

- Phishing emails
 - PowerShell execution
 - RDP access abuse
 - Credential dumping
-

3.3 Procedures (Real attacker behavior)

These are real-world implementations:

- Specific command lines
 - Malware behavior
 - Actual attack chains observed in incidents
-

4. Why MITRE ATT&CK is Important in SOC

4.1 Standardized Communication

SOC teams, incident responders, and red teams use the same language (e.g., T1059 instead of vague descriptions).

4.2 Better Detection Engineering

Helps SOC analysts build detection rules based on attacker behavior, not random alerts.

4.3 Improved Threat Hunting

Analysts can hunt systematically across techniques like lateral movement or credential access.

4.4 Faster Incident Response

You can quickly understand:

- Attack stage
- Likely next steps

- Required containment actions
-

4.5 Detection Gap Identification

Helps identify blind spots in SIEM coverage and security monitoring.

5. Key Factors SOC Analysts Must Focus On

5.1 Behavioral Analysis

Focus on patterns instead of single indicators like IPs or hashes.

5.2 Attack Chain Understanding

Connect events together:

- Phishing → Login → Privilege Escalation → Data Access
-

5.3 Technique Mapping

Every alert should be mapped to a MITRE technique. Example:

- PowerShell encoded command → Execution (T1059)
-

5.4 Data Source Coverage

Understand which logs map to which techniques:

- Authentication logs → credential attacks
 - Endpoint logs → malware execution
 - Network logs → C2 communication
-

5.5 Attack Progression Awareness

Understand attacker lifecycle stages:

- Reconnaissance
- Initial Access
- Persistence
- Exfiltration

6. How MITRE Improves SOC Skills

6.1 Threat Hunting Improvement

You move from reactive alert handling to proactive hunting.

6.2 Better SIEM Querying

You build structured queries based on attack techniques.

6.3 Faster Incident Investigation

You immediately understand attack context and severity.

6.4 Detection Engineering Skills

You start creating rules based on behavior patterns.

6.5 Professional Communication

You explain incidents clearly using standardized terminology.

7. Simple SOC Example

Scenario:

Multiple failed logins followed by a successful login.

Without MITRE:

- “Suspicious login activity detected”

With MITRE:

- Technique: Credential Access (Brute Force)
 - Tactic: Initial Access
 - Action: Investigate account + check lateral movement
-

8. Intermediate MITRE ATT&CK for SOC Analysts

At the intermediate level, SOC analysts move from simply understanding MITRE ATT&CK to **actively applying it in investigations, threat hunting, and SIEM detections.**

8.1 Intermediate Skill Focus Areas

8.1.1 Technique-Based Investigation

Instead of starting with alerts, you start with techniques:

- T1059 → Command Execution (PowerShell, CMD abuse)
- T1110 → Brute Force attacks
- T1021 → Remote Services (RDP, SMB)

You investigate logs based on behavior patterns tied to these techniques.

8.1.2 Mapping SIEM Data to MITRE

You begin aligning log sources with attack techniques:

- Authentication logs → Credential Access (T1110)
 - Windows event logs → Execution (T1059)
 - Network logs → Command & Control (T1071)
 - DNS logs → C2 or tunneling behavior
-

8.1.3 Intermediate Threat Hunting with MITRE

You start hunting proactively using MITRE techniques:

Example hunts:

- Search for repeated login failures (T1110)
 - Detect PowerShell encoded commands (T1059)
 - Identify unusual lateral movement (T1021)
-

8.2 Intermediate SPL + MITRE Mapping Examples

8.2.1 Brute Force Detection (T1110)

```
index=auth_logs action=failure  
| stats count by src_ip, user  
| where count > 10
```

8.2.2 Suspicious PowerShell Execution (T1059)

```
index=windows_logs "powershell" "-enc"  
| stats count by user, src_ip
```

8.2.3 Lateral Movement Detection (T1021)

```
index=auth_logs  
| stats dc(dest_host) as hosts by src_user  
| where hosts > 5
```

8.3 Intermediate SOC Investigation Workflow Using MITRE

1. Detect anomaly or alert
 2. Identify MITRE technique
 3. Expand scope (search related logs)
 4. Correlate multiple data sources
 5. Reconstruct attack chain
 6. Validate impact
 7. Document findings using MITRE mapping
-

8.4 Building MITRE-Based Dashboards

At intermediate level, SOC analysts build dashboards like:

- Top brute force attempts (T1110)
 - Top PowerShell execution events (T1059)
 - Lateral movement attempts (T1021)
 - Suspicious DNS activity (T1071)
-

8.5 Intermediate Attack Chain Understanding

You begin connecting attacker behavior:

Example chain: 1. T1110 → Brute force login 2. T1059 → Command execution after login 3. T1021 → Lateral movement 4. T1041 → Data exfiltration

8.6 Common Intermediate Mistakes

- Treating MITRE as theory instead of practice
 - Not mapping logs to techniques
 - Ignoring correlation between events
 - Focusing only on alerts instead of behavior
-

8.7 Skills Gained at Intermediate Level

- MITRE technique identification in logs
 - SIEM-based attack mapping
 - Threat hunting using behavioral patterns
 - Multi-source correlation analysis
 - Basic detection engineering mindset
-

9. Advanced MITRE ATT&CK for SOC Analysts

At the advanced level, SOC analysts transition into **detection engineering, automation, and adversary simulation using MITRE ATT&CK**. The focus is no longer only identifying techniques but **building systems that detect and respond to them automatically**.

9.1 Advanced Detection Engineering with MITRE

At this stage, analysts build detections directly mapped to ATT&CK techniques:

Examples:

- T1059 (Command Execution) → Detect encoded PowerShell + suspicious parent processes
- T1071 (C2 Communication) → Detect periodic beaconing patterns
- T1021 (Lateral Movement) → Detect unusual remote service usage
- T1003 (Credential Dumping) → Detect LSASS access or credential extraction behavior

9.2 Behavioral Analytics and Anomaly Detection

Advanced SOC teams move from rule-based detection to behavioral analytics:

- Establish baseline user and system behavior
- Detect deviations from normal activity
- Identify low-and-slow attacks

Example behaviors:

- Login from new country + unusual time
 - Rare process execution on critical servers
 - Unexpected outbound traffic spikes
-

9.3 MITRE-Based Threat Hunting at Scale

Advanced hunting is automated and continuous:

- Continuous queries mapped to ATT&CK techniques
- Scheduled hunts across SIEM data
- Cross-domain correlation (endpoint + network + cloud)

Example hunts:

- Persistent PowerShell usage across endpoints
 - Cross-host authentication anomalies
 - DNS tunneling detection patterns
-

9.4 Automation and SOAR Integration

MITRE ATT&CK becomes the foundation for automation:

Workflow example: 1. Detection triggers (T1110 brute force) 2. SOAR enriches IP + user context 3. AI evaluates risk score 4. Automated response (block IP / disable account) 5. Incident ticket created automatically

9.5 AI-Powered MITRE Mapping

Advanced SOC environments use AI to:

- Automatically map logs to ATT&CK techniques

- Summarize attack chains
- Predict next attacker move
- Generate incident reports

Example: Raw logs → AI → “This matches T1021 lateral movement + T1059 execution chain”

9.6 Adversary Emulation and Purple Teaming

SOC analysts collaborate with red teams:

- Simulate real MITRE techniques using tools like Caldera
- Validate detection coverage
- Improve SIEM rules based on attack simulation

Focus areas:

- Attack replay testing
 - Detection gap analysis
 - Control validation
-

9.7 Detection Coverage Management

Advanced SOC teams measure:

- % of MITRE techniques covered
- Which tactics are weakly monitored
- Detection effectiveness per technique

Example:

- 80% coverage of Execution techniques
 - 40% coverage of Lateral Movement techniques
-

9.8 Attack Chain Reconstruction (Advanced Level)

You reconstruct full attacker lifecycle using MITRE:

Example chain: 1. T1566 → Phishing 2. T1059 → Execution 3. T1071 → C2 communication 4. T1021 → Lateral movement 5. T1003 → Credential dumping 6. T1041 → Data exfiltration

9.9 Advanced SOC Use Cases

- Detect nation-state APT behavior
 - Identify fileless malware campaigns
 - Monitor cloud account takeover attempts
 - Detect insider threat behavior patterns
-

9.10 Skills at Advanced Level

- Detection engineering (Sigma/SPL/queries)
 - SOAR automation design
 - Threat hunting at scale
 - AI-assisted SOC operations
 - MITRE ATT&CK coverage mapping
-

10. Conclusion

MITRE ATT&CK is a core SOC framework that transforms analysts into proactive defenders. It improves threat hunting, detection engineering, and incident response by focusing on attacker behavior instead of isolated alerts.